PROFIT DIAGNOSTIC · CASE STUDY

Sales grew 29.8%. Gross profit fell $1.83M.

Where Adventure Works' reseller channel makes and loses money, why margin collapsed, and what to change in pricing and promotions.

> *"Our sales grew this year but profit didn't. We want to know where we make money, where we lose it, why margin changed, and what to change in pricing and discounting."*
>
> — *client brief*

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Prepared by
Saad Attia, Data & BI Analyst

Data
Microsoft AdventureWorksDW2022 reseller sales: public sample data for a fictional company

Period compared
Dec 2011–Nov 2012 (P0) vs Dec 2012–Nov 2013 (P1)

1 · EXECUTIVE SUMMARY

The growth came from products sold below cost

Reseller sales rose from $27.9M to $36.2M, but gross profit went from +$1.15M to −$0.67M. The cause is not discounting: it is a dealer price list that sets 13 product models below their standard cost, plus a new Touring range launched on that basis.

NET SALES, P1	GROSS PROFIT, P1	GROSS MARGIN, P1	SALES PRICED BELOW COST
$36.24M	**−$0.67M**	**−1.86%**	**60.5%**
+29.8% vs P0	P0: +$1.15M	P0: 4.13%	P0: 47.9%

Four findings

#	Finding	Impact (P1)
1	**The price list loses money before any discount.** 13 models have a dealer price below standard cost (Road −6.07%, Touring −3.60%, two jerseys and a cap −28.33%, three road frames −1.13%). They are 60.5% of sales.	−$2.04M
2	**New products were launched at a loss.** 118 SKUs new in P1 brought $20.9M of sales and lost money as a group; Touring-1000 alone lost $0.83M.	−$1.10M
3	**Costs rose faster than prices** on the 49 products sold in both periods: prices added $1.55M, costs took $2.00M.	−$0.45M
4	**Promotions deepen the loss.** The 7.6% of lines on promotion lost $1.07M; two promotions discount products that already sell below cost, and on promoted lines the price is reduced twice.	−$1.07M

Recommendations

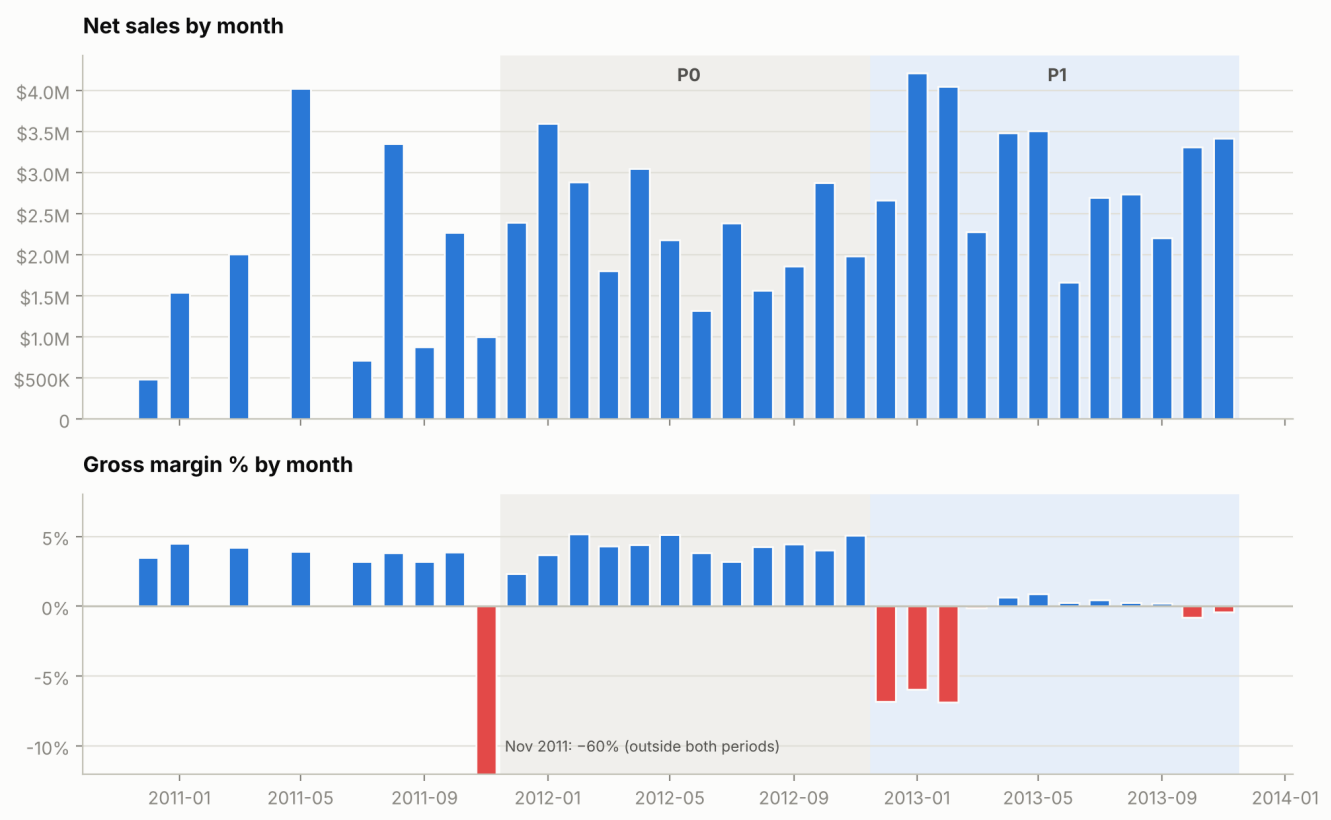
Action	Owner	Priority	Gross profit impact
Set a price floor: no dealer price below standard cost	Pricing / Finance	High	+$1.17M to +$1.70M
Stop promotions on below-cost products and apply each promotion once	Sales Director	High	up to +$0.64M
Pass standard-cost increases into the price list each quarter	Pricing / Finance	Medium	up to +$0.45M
Margin gate for new launches: no SKU listed below target margin	Product / Commercial	Medium	prevents recurrence

Impacts overlap, so they are not additive: the price floor together with removing line discounts is worth +$2.17M (volume held). The price-floor range runs from "every affected order is lost" (+$1.17M) to "volume held" (+$1.70M). A discount cap, the usual first idea, is worth only +$0.05M.

2 · WHAT HAPPENED

Margin turned negative in December 2012 and never recovered

Monthly gross margin held at roughly 2–5% through November 2012. In December 2012, the month the Touring range was first sold, it dropped to −6.8%, reached −6.9% in February 2013, and has hovered around zero since. Sales kept growing throughout, which is why the problem was easy to miss.



Monthly net sales and gross margin %. Shaded: the two 12-month periods compared. November 2011 (−60%) is a one-off outside both periods.

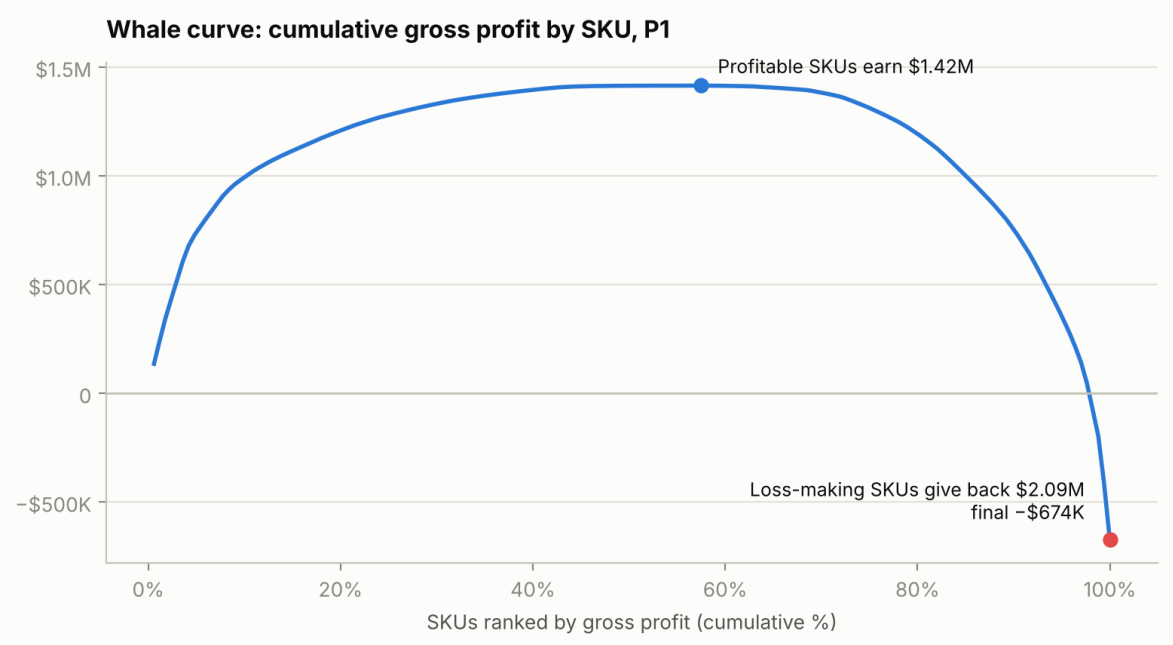
Period	Order lines	Units	Net sales	Gross profit	Margin
P0 · Dec 2011–Nov 2012	21,670	78,959	$27,921,671	$1,153,445	4.13%
P1 · Dec 2012–Nov 2013	30,726	113,529	$36,240,485	−$673,727	−1.86%
Change	**+41.8%**	**+43.8%**	**+29.8%**	**−$1,827,172**	**−5.99 pts**

Why these periods. The data ends in November 2013 and 2011 has three months with no orders, so calendar years are not comparable. Two complete 12-month windows are.

3 · WHERE PROFIT IS MADE (Q1)

A handful of products earn the profit; most of the volume gives it back

Category, P1	Net sales	Share	Gross profit	Margin	Margin P0
Bikes	$28,866,428	79.7%	−$1,232,916	−4.27%	1.64%
Components	$6,003,210	16.6%	$354,645	5.91%	12.40%
Clothing	$960,797	2.7%	$59,987	6.24%	23.46%
Accessories	$410,050	1.1%	$144,558	35.25%	29.23%



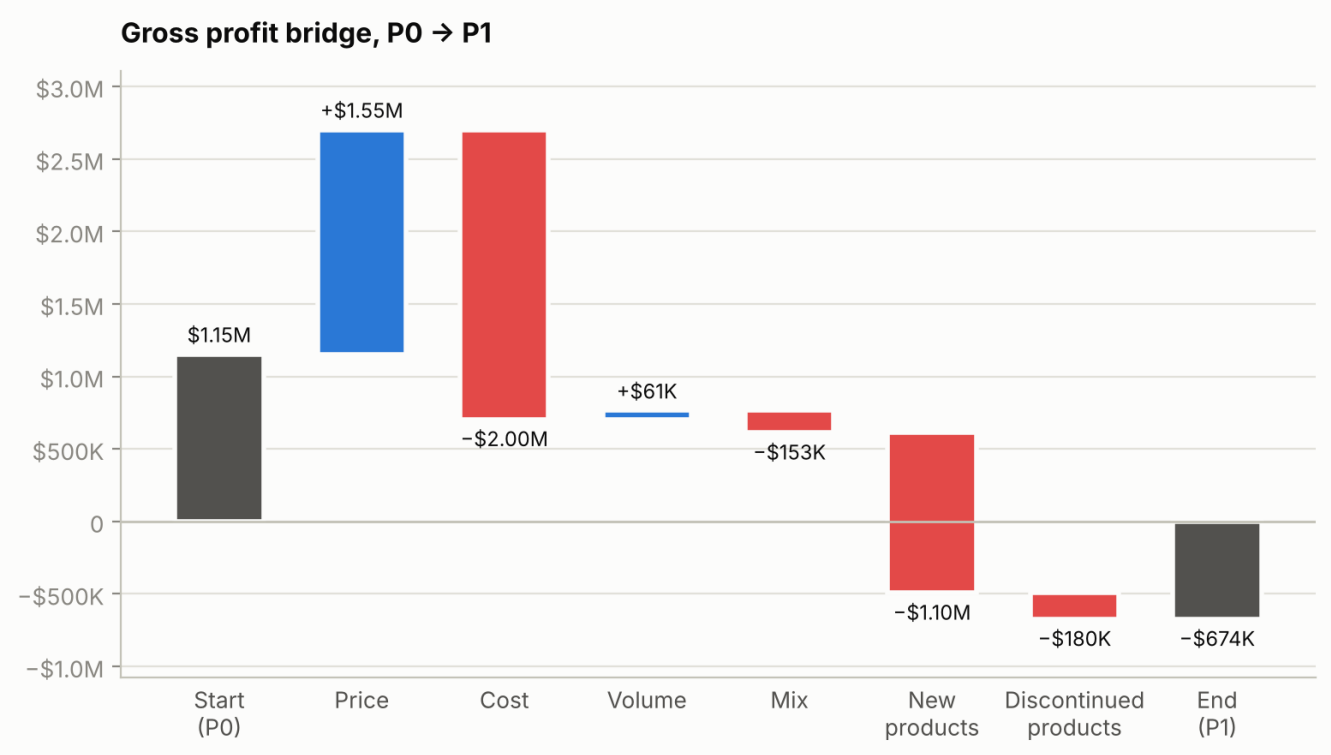
SKUs ranked from most to least profitable. The curve peaks at $1.42M, then loss-making SKUs pull it down to −$0.67M.

- **26 SKUs (16% of 167)** earn 80% of the profit made by profitable SKUs; Mountain-200 alone earns $0.61M.
- **71 SKUs lose money** and carry 65.5% of sales.
- **Worst models:** Touring-1000 −$0.83M, Touring-3000 −$0.34M, Road-350-W −$0.29M, Road-250 −$0.20M.
- **297 of 493 resellers** are unprofitable, holding 66.8% of sales: they buy the loss-making range, not because they negotiate harder.
- **Not a sales-team problem:** discount rates range only 0.05%–3.24% across reps, and 19 of 30 rep-territory pairs lose money on the same price list.

4 · WHY PROFIT CHANGED (Q2)

New products and rising costs explain the $1.83M swing

A price-volume-mix bridge splits the change in gross profit into its causes at SKU level. Only 49 of the 167 SKUs sold in P1 were also sold in P0; the rest of the range was replaced, so new and discontinued products are shown separately.

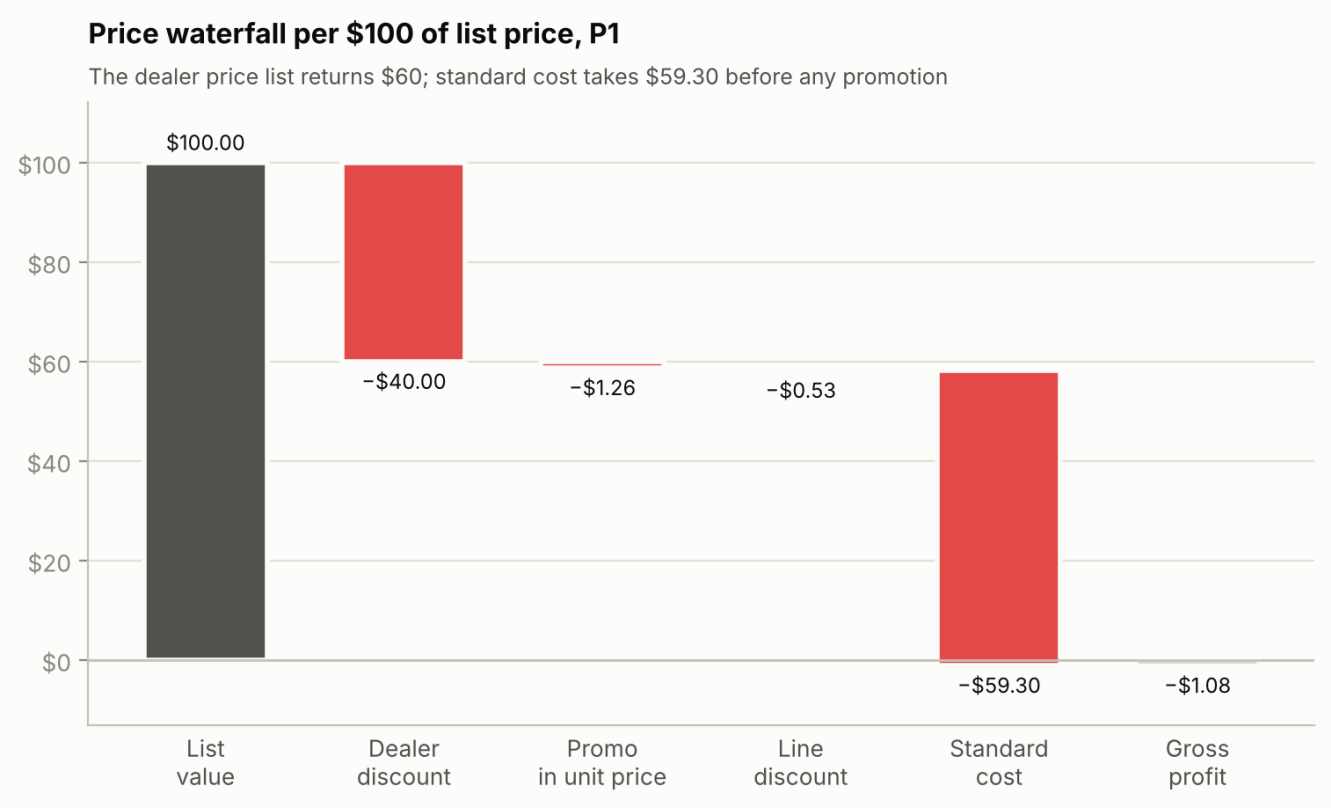


Driver	Gross profit	What it means
Price	+$1,548,052	Continuing products were sold at higher net prices.
Cost	−$1,997,686	Their standard cost rose more than price: a net squeeze of $0.45M (Road-250 −$0.17M, Road-550-W −$0.10M).
Volume	+$61,087	More units at P0 margins added little, because P0 margins were already thin.
Mix	−$153,341	Within continuing products, the mix moved to lower-margin items.
New products	−$1,104,857	Touring-1000/3000/2000, Road-350-W and Road-750 launched below cost.
Discontinued	−$180,427	Products dropped after P0 had been profitable.
Total change	**−$1,827,172**	**Reconciles exactly to the period totals.**

On net sales, the same bridge shows +$20.9M from new products and −$12.3M from discontinued ones: the growth story is a range replacement, not more of the same products.

5 · WHERE THE MONEY LEAKS (Q3)

The leak is in the price list, not in the discounts



Per $100 of list price, P1. Resellers pay the dealer price (60% of list); standard cost takes $59.30 of it. Promotions and line discounts cost only $1.79 together.

Products priced below cost

Model group	Dealer price vs cost	Gross profit P1
Touring-1000/2000/3000	−3.60%	−$1,237,595
Road-250/350-W/550-W/750	−6.07%	−$658,840
Two jerseys, cycling cap	−28.33%	−$128,912
Three road frames	−1.13%	−$16,890
All 13 models	**60.5% of sales**	**−$2,042,237**

Promotions on top

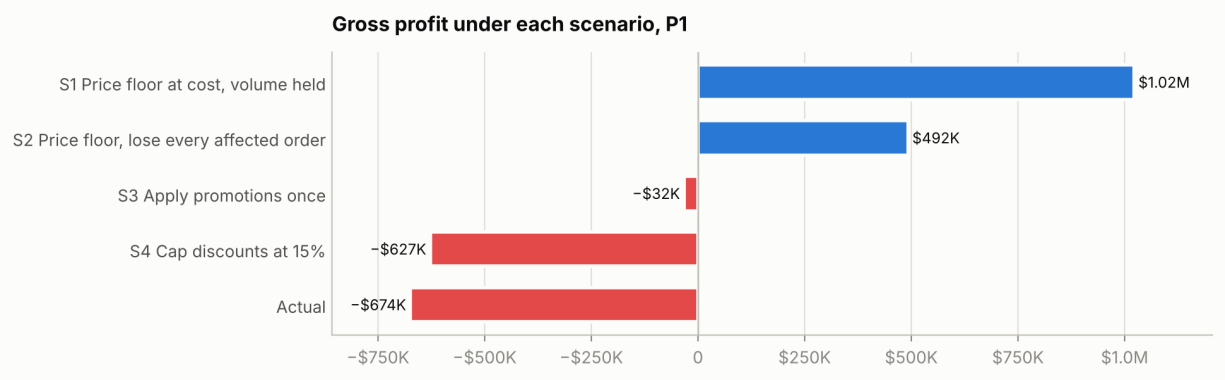
Discount band	Lines	Margin
No discount	28,388	1.20%
0–10%	1,349	−5.93%
10–20%	820	−79.70%
30%+	169	−354.61%

Touring-1000 promotion (20%) lost $547,905 and Touring-3000 promotion (15%) $277,071: both discount products already priced below cost.

Data finding to confirm with the client. On promoted lines the invoiced unit price is already below the dealer price by the promotion rate × list price, and the promotion's line discount is then applied again. If the promotion was meant to be applied once, this costs $0.78M a year in the unit price alone.

6 · WHAT TO DO (Q4)

Fix the price list first; promotions second



Gross profit in P1 if each change had been in place. Volume held constant unless stated; S2 assumes every order containing a below-cost line is lost.

1 · Set a price floor: no dealer price below standard cost

High · +$1.17M to +$1.70M

Observation	13 models carry a dealer price below standard cost; they are 60.5% of P1 sales.
Evidence	Below-cost lines lost $2.04M while all other lines earned $1.37M.
Action	Raise dealer prices to at least cost: Road +6.07%, Touring +3.60%, road frames +1.13%. Jerseys and cap need +28.33% or a cheaper supplier.
Why it is safe	Orders containing below-cost lines lose $1.17M even after their profitable items, so losing every one of them would still raise profit by $1.17M.
Expected outcome	Gross profit between +$0.49M and +$1.02M on the P1 base; margin back above zero.

2 · Stop promotions on below-cost products; apply each promotion once

High · up to +$0.64M

Evidence	Promoted lines (7.6% of lines) lost $1.07M; the two Touring promotions alone lost $0.82M.
Action	Suspend Touring promotions now; confirm with billing whether promotions are applied twice and correct the price rule.
Expected outcome	+$0.64M if promotions are applied once. Combined with the price floor and no line discounts: +$2.17M, because Touring-1000 still loses $0.25M after the floor while its 20% promotion stays in place.

3 · Pass cost increases into the price list each quarter

Medium · up to +$0.45M

Evidence	On continuing products, costs took $2.00M while prices added $1.55M.
Action	Review dealer prices whenever standard cost changes; flag any SKU whose margin at dealer price falls below target.

4 · Add a margin gate to new product launches

Medium · prevents recurrence

Evidence	New products lost $1.10M in their first year; Touring was launched below cost.
Action	No SKU enters the price list without a dealer-price margin check signed off by Finance.

Not recommended as a first step: a discount cap at 15% adds only $47K, because discounts are not where the money goes.

7 · QUESTIONS FOR THE CLIENT

- **Is standard cost current and complete?** Every finding uses standard cost; if actual cost is lower for Touring or Road models, the loss shrinks.
- **Is the double promotion reduction intended?** (Section 5.) If not, it is a billing rule to fix immediately.
- **Why were Touring dealer prices set below cost?** A deliberate market-entry decision needs an end date and a target margin.
- **What is given off-invoice?** Rebates, payment terms and free freight are not in the data; the true pocket margin is likely lower.

8 · METHOD, DATA QUALITY AND LIMITS

Method

- **Q1** Profit by category, model, SKU, reseller and rep; ABC and whale curve on gross profit.
- **Q2** Price-volume-mix bridge at SKU level, with a separate cost effect and new/discontinued buckets. Effects reconcile to the cent.
- **Q3** Price waterfall from list price to gross profit; discount bands; below-cost pricing test.
- **Q4** Line-level repricing scenarios with an upper bound (volume held) and a lower bound (affected orders lost).
- Margins are always ratios of totals, never averages of line percentages.

Data quality

- 60,855 order lines; one row per line confirmed; no nulls, no orphan keys.
- Price × quantity, discount and cost arithmetic reconcile on every line.
- Amounts are in USD across all five currency codes (same product, same price).
- SKU used as product key, because product keys change when price or cost changes.

Limits

- Public sample data for a fictional company: the findings demonstrate the method.
- Gross profit only: freight, overheads and off-invoice items are out of scope.
- Scenarios do not model price elasticity beyond the two bounds.

WHAT THE CLIENT RECEIVES

Deliverable	What it does for the business
Data-quality log and audit scripts	Shows exactly what was checked, so numbers can be trusted in a board meeting.
SQL analysis layer (views and six scripts)	Re-runs the whole diagnostic on next quarter's data in minutes.
Power BI pricing simulator	Lets the commercial team set a price floor and a discount cap and see the profit, the price increase needed and the affected models before changing the price list.
This report: findings, $ impact, actions	Turns the analysis into four decisions with owners and priorities.
Monitoring KPIs	Share of sales priced below cost, price realization, gross margin by model: one look per month catches the next leak early.